

CA1: Mini Project – Project Proposal & System Design (Phase 1)

NLP Mini Project – Team Proposal Report

An Integrated Pipeline for Fraudulent Job Filtering, Seniority Classification, and Explainable Candidate-Job Matching

LinkedIn Job Posting Classification and Personalized Job Recommendation System

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1 Title and Objective

Title

An Integrated Pipeline for Fraudulent Job Filtering, Seniority Classification, and Explainable Candidate-Job Matching – proposed as a LinkedIn Job Posting Classification and Personalized Job Recommendation System.

Objective

The objective of this project is to design a system that (1) filters fraudulent job postings before they propagate into downstream processes, (2) classifies genuine postings by category and seniority level, and (3) recommends ranked, explainable job matches to candidates based on résumé-to-requirement similarity. This directly addresses a gap identified in our literature survey (Section 3), where fraud filtering, classification, and recommendation are typically designed, trained, and evaluated as three disconnected systems rather than a single coherent pipeline. This report documents that design in full: the problem it addresses, the literature that motivates it, the datasets it will draw on, the preprocessing and modeling approach it proposes, and the team plan for delivering it.

Motivation

Online recruitment has become the dominant channel through which job seekers discover opportunities, but that scale brings two well-documented costs. First, the same openness that makes large job boards useful also makes them a target for fraudulent postings – listings designed to harvest personal information, extract fees, or simply inflate a platform’s apparent activity. Second, even among genuine postings, the sheer volume makes manual browsing an increasingly poor way for a candidate to find roles that actually match their skills and experience. Most existing systems address only one side of this problem at a time: fraud-detection research rarely considers what happens downstream in a recommendation pipeline, and recommendation research typically assumes an already-clean posting pool. This project proposes to address both sides together, and to treat the ordering between them – filter first, then classify, then recommend – as a deliberate design decision rather than an afterthought.

2 Problem Statement

Online professional networks such as LinkedIn host a continuously growing volume of job postings spanning industries, roles, and experience levels. Two related challenges arise for both platforms and job seekers:

1. Accurately classifying job postings by **category**, **seniority**, and **genuineness**, given the well-documented prevalence of fraudulent listings; and
2. **Matching candidates** to the postings most relevant to their skills, experience, and preferences.

Existing academic and applied work treats these as separate problems: fraud/genuineness detection is studied independently of category or seniority classification, and recommendation systems are built assuming an already-clean posting pool. In practice, fraudulent postings that leak into a recommendation pipeline directly degrade recommendation quality and erode user trust. This project proposes a single pipeline that filters, classifies, and recommends in sequence, so that each downstream stage operates only on postings that have already been validated by the stage before it.

Illustrative Scenario

Consider a recent graduate searching for entry-level data-analytics roles on a large job board. Without genuineness filtering, some fraction of the postings they encounter are scams designed to harvest per-

sonal information rather than genuine openings – a candidate has no reliable way to tell the difference from the listing text alone. Without seniority classification, postings mislabeled or unlabeled by the original poster are effectively invisible to a seniority-based search filter, even when they are genuinely relevant. Without explainable recommendation, even a technically accurate match gives the candidate no way to judge whether the system considered their actual skills or something else entirely. Each of this project’s three modules addresses one link in that chain; the system design in Section 5 treats the chain as a whole rather than three independent tools a candidate would need to use separately.

Scope of This Phase

This report covers Phase 1 of the project: problem framing, literature review, dataset selection, and system design. It proposes an approach and a plan; it does not report model training results, evaluation metrics, or system performance, since implementation and evaluation are the subject of the project’s next phase. Where this report describes a technique (a preprocessing step, a modeling choice), it describes what is planned and why, not what has already been measured.

Success Criteria for This Phase

Consistent with that scope, success for Phase 1 is defined by planning quality rather than model performance: a clearly justified real-world problem (Section 1), a literature review that identifies a genuine, well-supported gap (Section 3.5), datasets that are relevant, adequately sized, and appropriately sourced for each module (Section 4), preprocessing techniques that are appropriate and clearly documented (Section 5.2), and a team plan with clearly defined roles and a realistic timeline (Section 8). Model accuracy targets are intentionally deferred to the project’s next phase, where they can be stated against actual training results rather than asserted in advance.

Deliverables for This Phase

Consistent with the assignment brief for this phase, the deliverable is this Project Proposal Report in PDF format, covering the five required sections (Title and Objective, Problem Statement, Literature Context, Dataset Details, Methodology and System Design) plus the team responsibilities and timeline documentation this rubric also requires. No code, trained models, or evaluation results are included in this deliverable, since implementation is scoped to the project’s next phase. Each team member’s individual contribution to this proposal is recorded in Section 8.

3 Literature Context

3.1 Job Posting Classification

Nasser and Alzaanin [1] evaluated five classical machine learning classifiers – Multinomial Naive Bayes, SVM, Decision Tree, KNN, and Random Forest – for separating genuine and fraudulent job postings using TF-IDF text features on the EMSCAD dataset, establishing a widely-cited baseline for classical-ML approaches to this task. A follow-up genuineness-classification study [2] applied Random Forest with an oversampling strategy to correct class imbalance in real-vs-fake postings on the same dataset, reporting accuracy above 98%. Both studies work from the same underlying premise that motivates our own Dataset choice for this sub-problem (Section 4.2): that TF-IDF text representations, paired with classical ML classifiers and an explicit strategy for class imbalance, are a strong and reproducible starting point for fraud/genuineness detection.

More recent work reframes job-posting analysis as multi-class metadata extraction rather than binary fraud detection [3], using transformer embeddings (BERT/USE4) and LLM-generated synthetic data to offset limited labeled-data availability. This signals a plausible extension path beyond the classical

baselines: once a classical pipeline is working and evaluated, a transformer-based variant is a natural next comparison point, and this project’s design (Section 5) leaves room for that extension explicitly. Large-scale labour-market-intelligence research [4] has applied ML classification to job vacancies with an emphasis on multi-language, standardized taxonomies rather than proprietary single-language category systems, enabling broader real-time market monitoring – a design consideration worth noting given that this project’s chosen LinkedIn dataset (Section 4.1) uses its own internal skill and industry taxonomy, whose granularity we examine closely during dataset planning.

3.2 Preprocessing and Feature Extraction Approaches in Prior Work

Since this proposal’s own preprocessing plan (Section 5.2) draws directly on established practice, it is worth reviewing what the surveyed literature actually does at the feature-extraction stage rather than only at the modeling stage. The classical fraud-classification baselines [1, 2] both rely on TF-IDF vectorization over cleaned, lowercased, stopwords-filtered text – a comparatively lightweight pipeline chosen specifically so that the modeling comparison (Naive Bayes vs. SVM vs. Random Forest) is not confounded by an elaborate feature-engineering step. The more recent metadata-extraction work [3] moves to transformer embeddings (BERT, Universal Sentence Encoder) precisely because TF-IDF’s bag-of-words assumption discards word order and struggles with synonymy – e.g. “machine learning engineer” and “ML engineer” produce near-orthogonal TF-IDF vectors despite being near-synonymous job titles. This tradeoff is directly relevant to our own plan: TF-IDF is proposed as the baseline representation for reproducibility and interpretability, with a transformer embedding representation planned as a direct comparison once the baseline is validated, rather than choosing one representation permanently at the proposal stage.

On the recommendation side, DSSM-style approaches [5] similarly move past raw keyword matching toward learned semantic representations, and the hybrid-filtering literature [7] combines content-based similarity (which needs a good text representation) with collaborative signals (which this project’s datasets do not directly support, since we do not have historical candidate-application interaction data). This is a scoping decision worth stating plainly: this project’s recommendation module is planned as content-based, not collaborative, because the available dataset (Section 4.3) supports the former and not the latter.

3.3 Class Imbalance Handling in Prior Work

Class imbalance is a recurring theme across the fraud-detection literature specifically, since fraudulent postings are a small minority of any real posting pool by construction. The Random-Forest-with-oversampling study [2] addresses this directly, and its reported accuracy above 98% is a useful benchmark for our own planned EMSCAD baseline. More broadly, oversampling (SMOTE and its variants), class-weighting, and threshold adjustment are the three standard approaches to this problem in the classification literature; this proposal’s plan (Section 5.2) opts for SMOTE specifically for the binary, more strongly imbalanced Module 1 problem, and class-weighting for the six-class, less extremely imbalanced Module 2 problem, on the reasoning that SMOTE’s synthetic-sample generation is more justified when the minority class is a very small fraction of the data, while class-weighting is a cheaper and sufficient correction when the imbalance, though real, is less severe.

Mishra and Rathi [5] proposed an enhanced Deep Semantic Structured Model (DSSM) to improve semantic matching between résumés and job descriptions beyond earlier keyword-based techniques. Deep reinforcement learning has also been applied to job recommendation [6], using historical user-profile and job-description data to outperform traditional recommendation baselines in experimental evaluation. Hybrid filtering – combining collaborative and content-based approaches – has been used to improve precision, recall, and F1 in systems that recommend careers based on candidate skills and interests [7].

Personality-aware matching approaches [8] extend recommendation beyond hard-skill overlap, integrat-

ing trait-based assessments with LLM-processed job descriptions to account for career fit and retention, not just qualification match. CareerCue [9] proposes a modular pipeline that parses LinkedIn profile data for skills and experience, fetches live postings via a job-aggregation API, and produces ranked, explainable recommendations – while explicitly noting platform dependency, limited personalization, and data-privacy concerns as unresolved issues. A broader review of ML-based job recommendation systems [10] documents the shift toward hybrid recommendation models and highlights fairness and debiasing in ranking as an active research concern. Taken together, this thread of literature converges on semantic, content-based matching (as opposed to pure keyword overlap) as the current state of the art, and on explainability as a persistently under-addressed requirement – both of which directly shape the recommendation module’s design in Section 5.5.

3.4 Existing Solution Analysis

LinkedIn’s own platform already performs large-scale job matching (its “Jobs for You” feed), but the underlying ranking models and signals are proprietary and closed to academic study – leaving room for an independent, transparent system built on public job-posting data. Academic prototypes such as résumé-based company recommenders demonstrate the feasibility of a résumé → feature extraction → classifier/recommender pipeline (KNN, SVM, Random Forest), but are typically validated on small, single-institution datasets rather than a real job board’s actual posting corpus. CareerCue-style systems show that a profile extractor plus job fetcher plus scoring/notification layer is achievable with lightweight tooling, but generally rely on rule-based scoring rather than transformer-based semantic matching, and are rarely benchmarked against classical ML baselines.

Approach	Strength	Limitation relative to this proposal
LinkedIn “Jobs for You”	Massive scale, real user engagement signal	Proprietary, not reproducible or auditable
Classical fraud classifiers [1, 2]	Strong, reproducible accuracy on EM-SCAD	Evaluated in isolation from recommendation
DSSM / RL recommenders [5, 6]	Semantic matching beyond keyword overlap	No fraud-filtering stage upstream
CareerCue [9]	End-to-end profile-to-recommendation pipeline	Rule-based scoring, limited explainability

Table 1: Summary comparison of surveyed approaches against this project’s proposed scope.

Across the systems surveyed, two gaps recur, summarized in Table 1. First, job-posting genuineness/fraud filtering and job recommendation are almost always treated as separate problems, despite fraudulent postings directly degrading recommendation quality when left unfiltered. Second, explainability is inconsistently addressed – most systems return a ranked list without surfacing why a job was matched, which limits user trust, as noted explicitly in the CareerCue evaluation.

3.5 Identified Gap and Justification

The reviewed literature treats classification (category/seniority/genuineness) and recommendation (candidate-job matching) as largely independent problems, each with mature but separate baselines. A combined pipeline – filtering and categorizing postings before they enter a hybrid, explainable recommendation stage – addresses gaps common to both threads: fraudulent-listing leakage into recommendations, weak explainability, and limited handling of cold-start users or newly posted jobs. This is the specific contribution this project proposes to make: not a new classification or recommendation algorithm in isolation, but a system-level integration that the surveyed literature does not attempt.

4 Dataset Details

The project will draw on three complementary, publicly available datasets, one per pipeline stage. All three have been identified and reviewed during this proposal phase; collection and preprocessing are planned for the project’s next phase.

4.1 LinkedIn Job Postings (2023–2024) — Category/Seniority Classification

Source: Kaggle, dataset *arshkon/linkedin-job-postings*. 123,849 real LinkedIn postings scraped across industries and locations, split across 11 files: a core `postings` table plus normalized `skills`, `industries`, `benefits`, and `salaries` tables keyed by `job_id`, and `companies` metadata.

File	Rows	Purpose
<code>postings.csv</code>	123,849	Core table: title, description, location, salary fields, <code>formatted_experience_level</code> , work type
<code>jobs/job_skills.csv</code>	213,768	<code>job_id</code> → <code>skill_abr</code> (many-to-many)
<code>jobs/job_industries.csv</code>	164,808	<code>job_id</code> → <code>industry_id</code>
<code>jobs/benefits.csv</code>	67,943	<code>job_id</code> → benefit type
<code>jobs/salaries.csv</code>	40,785	Normalized salary records
<code>mappings/skills.csv</code>	35	<code>skill_abr</code> → <code>skill_name</code>
<code>mappings/industries.csv</code>	422	<code>industry_id</code> → <code>industry_name</code>
<code>companies/companies.csv</code>	24,473	<code>company_id</code> → name, size, location

Planned classification target: `formatted_experience_level` – six classes (Entry, Associate, Mid-Senior, Director, Executive, Internship). Initial exploration shows this field populated for roughly three-quarters of postings, with the remainder forming a pool the trained model would need to label at inference time rather than discard – a design requirement carried into Section 5.4. **Anticipated data-quality considerations:** salary fields appear sparsely populated across the dataset and are not planned as a primary feature for this reason. The skills/industries mapping tables use LinkedIn’s own internal taxonomy rather than a free-form skill vocabulary – notably, the skills mapping table has only 35 entries total, which initial review flags as a possible granularity concern worth close attention during preprocessing (Section 5.2), since an overly coarse taxonomy would limit its usefulness for fine-grained matching in Module 3.

4.2 EMSCAD — Genuineness/Fraud Classification

Source: Kaggle, dataset *shivamb/real-or-fake-fake-jobposting-prediction* (Employment Scam Aegean Dataset). 17,880 job postings published 2012–2014, of which 866 (4.8%) are manually annotated as fraudulent – the same dataset used by the classical-ML baselines in Section 3.1, which will let our planned reproduction be compared directly against published results.

Fields: title, location, department, salary_range, company_profile, description, requirements, benefits, telecommuting, has_company_logo, has_questions, employment_type, required_experience, required_education, industry, function, fraudulent (target). **Anticipated data-quality considerations:** the dataset is strongly imbalanced (17,014 genuine vs. 866 fraudulent postings), which the methodology plan (Section 5.3) addresses with SMOTE oversampling applied only to a training split, to avoid inflating any future evaluation with synthetic samples.

4.3 Résumé–Job Matching Dataset — Recommendation Module

Source: Kaggle résumé-parsing/job-matching dataset. 9,544 paired records, each combining a candidate résumé (career objective, skills, education, work history, certifications) with a target job’s requirements (position, required education, required experience, required skills) and a pre-computed `matched_score`.

This is the key asset for Module 3: it supplies *ground-truth* match scores, so the recommendation component can be trained as a supervised ranking/regression task rather than relying purely on unsupervised similarity heuristics, which is a step beyond most of the surveyed recommendation literature. **Anticipated data-quality considerations:** initial inspection shows that structurally similar fields are serialized differently across columns (list-formatted vs. plain delimited text), which the preprocessing plan treats as a parsing requirement rather than something to defer.

4.4 Dataset Summary

Table 2 summarizes all three datasets against the pipeline stage they will support.

Dataset	Size	Target Module	Prediction Target
LinkedIn Job Postings	123,849 rows	Module 2 (Category/Seniority)	<code>formatted_experience_level</code> (6 classes)
EMSCAD	17,880 rows	Module 1 (Genuineness)	<code>fraudulent</code> (binary)
Résumé–Job Matching	9,544 pairs	Module 3 (Recommendation)	<code>matched_score</code> (continuous)

Table 2: Summary of the three datasets proposed for this project, one per module.

5 Methodology and System Design

5.1 Design Rationale

The system is proposed as three modules wired into a single sequential pipeline (Figure 1). Postings would first pass through the genuineness filter (Module 1); only postings scored genuine would be labeled by the category/seniority classifier (Module 2) and become eligible to enter the candidate-matching pool (Module 3). This ordering – filter, then classify, then recommend – is the project’s central architectural decision, directly motivated by the literature gap identified in Section 3.5: fraudulent postings should never reach a recommendation stage undetected. Each module is designed to be trained and validated independently first, so that its own performance can be measured cleanly before the modules are combined – and only then evaluated together as an integrated system, since a module’s in-isolation performance is not automatically evidence that it will behave well once wired into the full pipeline.

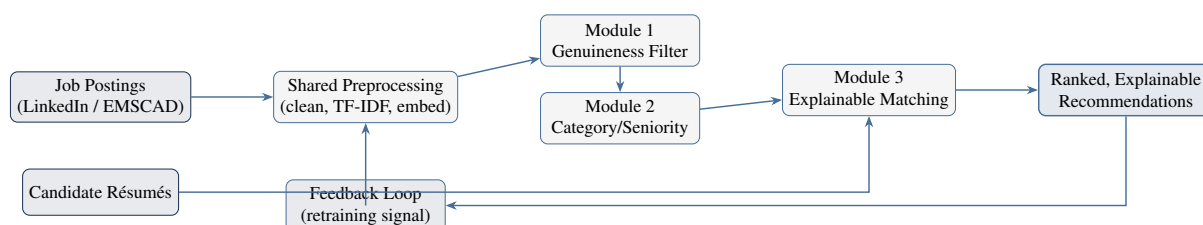


Figure 1: Proposed system architecture: sequential fraud filtering and classification gate the pool of postings eligible for explainable candidate matching, with a feedback loop for continuous retraining.

5.2 Data Preprocessing Plan

Preprocessing is planned as a shared pipeline applied consistently across all three datasets' free-text fields (`description`, `career_objective`, `responsibilities`, `requirements`), so that no module's text representation is built with different assumptions than the others':

- **Lowercasing and noise removal:** all text lowercased; HTML tags and URLs stripped via pattern matching, since several source fields (job descriptions in particular) are known to contain markup artifacts from web scraping.
- **Tokenization:** whitespace/regex-based tokenization after non-alphabetic character removal, chosen over a heavier subword tokenizer at this stage to keep the classical-ML baselines (Section 3.1) directly comparable to their published counterparts.
- **Stopword removal:** standard English stopwords list, to reduce dimensionality in the resulting TF-IDF representation without discarding domain-relevant terms.
- **Normalization:** lemmatization rather than stemming, to keep cleaned tokens as recognizable dictionary words – relevant later for the explainability goal in Module 3, where surfaced feature importances need to be human-readable.
- **Structured/categorical fields:** binary flags (e.g. `telecommuting`, `has_company_logo`) planned as direct numeric features alongside the text representation, rather than being discarded or text-encoded, since the literature identifies them as informative signals in their own right.
- **Class imbalance:** SMOTE oversampling planned for Module 1's training split specifically, given EMSCAD's 4.8% fraud rate; Module 2's six-class imbalance is planned to be handled via class-weighted loss instead, since oversampling at that dataset's row count and feature dimensionality would be considerably more expensive.
- **Feature extraction:** TF-IDF representations (planned at 5,000 features, unigrams and bigrams) as the primary text representation across all three modules, consistent with the classical baselines reviewed in Section 3, with a transformer-embedding representation (e.g. SBERT-style sentence embeddings) planned as a comparison point once the classical pipeline is working end-to-end.

5.3 Module 1 — Genuineness Classification (Planned Approach)

Trained on EMSCAD. Baseline plan: TF-IDF features (`title`, `description`, `requirements`) combined with the three structured binary flags described in Section 5.2, evaluated across Naive Bayes, SVM, and Random Forest classifiers, with SMOTE applied to the training split to address the 4.8% fraud rate, directly replicating the methodology of [1, 2] as a baseline before any extension. These three classifiers are planned specifically because they span a useful range of modeling assumptions: Naive Bayes as a fast, strongly-regularized baseline that assumes feature independence; linear SVM as a margin-based classifier well-suited to high-dimensional sparse TF-IDF input; and Random Forest as a non-linear ensemble capable of capturing feature interactions neither of the other two can. Training all three and comparing them empirically is preferred over committing to one in advance, since the literature itself does not agree on a single best classifier for this task. **Planned output:** a binary genuineness score attached to every posting, which becomes the gating signal for Module 2.

5.4 Module 2 — Category/Seniority Classification (Planned Approach)

Trained on the LinkedIn Postings dataset. Planned as a multi-class classifier predicting `formatted_experience_level` from `title`, `description`, and mapped skills/industries, using the same TF-IDF representation strategy as Module 1 for consistency. Class-weighted classical classifiers (Logistic Regression, SVM, Random Forest) are planned as the baseline comparison set, again chosen to compare a linear-probabilistic model, a margin-based model, and a non-linear ensemble against each other rather than assuming in advance which style of classifier suits a six-class, imbalanced target best. Postings without a ground-truth label are planned to be scored by the trained model at

inference time rather than discarded, since roughly a quarter of the dataset falls into this category and discarding it would meaningfully shrink the usable posting pool for downstream recommendation.

5.5 Module 3 — Recommendation Engine (Planned Approach)

Trained on the résumé–job matching dataset. Planned features are the semantic similarity between candidate-side text (skills, career objective, responsibilities) and job-side text (required skills, education, experience), trained against the `matched_score` ground truth as a supervised regression target rather than an unsupervised similarity heuristic – a deliberate choice, since an unsupervised similarity score alone would have no way to be calibrated against this dataset’s actual notion of a good match, whereas a supervised model can learn that calibration directly from the provided ground truth. Feature-level contributions (e.g. which skills or education fields drove a given score) are planned to be surfaced alongside each recommendation, directly addressing the explainability gap identified in Section 3.4. Both a small set of named, interpretable similarity features and a higher-capacity representation are planned to be compared once training begins, so that the accuracy/explainability tradeoff this problem presents is decided with evidence rather than assumed in either direction at the proposal stage.

5.6 Evaluation Plan

Each module is planned to be evaluated with metrics matched to its task, on a held-out test split never seen during training or preprocessing-parameter selection:

- **Module 1 (Genuineness):** accuracy, precision/recall/F1 on the fraud class specifically (not just overall accuracy, since the class imbalance described in Section 4.2 makes overall accuracy a poor standalone signal), and ROC-AUC. Confusion matrices are planned to be reported alongside summary metrics so that the precision/recall tradeoff for each candidate classifier is visible, not just a single blended score.
- **Module 2 (Category/Seniority):** accuracy, macro-F1, and weighted-F1 across all six classes, with macro-F1 planned as the primary model-selection metric specifically because accuracy alone would reward a model for the dominant classes (Mid-Senior, Entry) while masking poor performance on rare classes (Executive, Internship).
- **Module 3 (Recommendation):** mean absolute error (MAE), root mean squared error (RMSE), and R^2 against the `matched_score` target, since this is planned as a regression task rather than a classification one.
- **Integrated pipeline:** beyond each module’s own metrics, the integration plan (Section 5.7) calls for inspecting a sample of end-to-end recommendations qualitatively – checking whether top-ranked results are topically sensible for a given candidate profile – since a strong module-level metric does not by itself guarantee that the combined system produces sensible output once modules are chained together.

For every module, multiple candidate algorithms (at minimum two, per this phase’s requirements) are planned to be trained and compared side by side rather than committing to a single algorithm in advance, so that model selection is evidence-based once training is underway.

5.7 Integration Plan

Once each module is independently trained and validated, the fusion plan is to route a sample of postings through the full sequence shown in Figure 1: Module 1 filters, Module 2 labels the survivors, and Module 3 ranks them against a candidate profile with an explanation attached to each ranked result. This integrated run is planned as a distinct evaluation step from each module’s individual validation, on the view that a module’s standalone metrics are not sufficient evidence that the combined system will behave as designed – a risk this proposal treats as worth planning for explicitly rather than assuming away.

6 Risk Assessment and Mitigation Strategy

A proposal that only describes what is expected to work is less useful than one that also names what could go wrong. Table 3 sets out the risks this team has identified during planning, alongside the mitigation each risk motivates in the design already described in Section 5.

Risk	Likelihood	Planned Mitigation
Severe class imbalance in EMSCAD (4.8% fraud) degrades Module 1 recall	High	SMOTE oversampling on the training split only (Section 5.2); evaluation plan reports precision/recall/F1 on the fraud class specifically, not just accuracy (Section 5.6)
LinkedIn’s skill taxonomy (35 categories) is too coarse for fine-grained matching in Module 3	Medium-High	Flagged explicitly during dataset review (Section 4.1); free-text description fields planned as a fallback signal alongside the mapped taxonomy
Models trained on one dataset (e.g. the résumé–job matching corpus) may not generalize when applied within a pipeline populated by a different dataset’s postings (LinkedIn)	Medium-High	Integration plan (Section 5.7) treats end-to-end pipeline evaluation as a distinct, required step rather than assuming module-level metrics transfer automatically
24% of LinkedIn postings lack a ground-truth seniority label	Certain (known from initial exploration)	Design requirement to score unlabeled postings at inference rather than discard them (Section 5.4)
Heterogeneous data formats across datasets (e.g. list-encoded vs. plain-text fields) slow down preprocessing	Medium	Identified during dataset review (Section 4.3) and scheduled explicitly in the project timeline (Table 4) rather than left as an unplanned blocker

Table 3: Risks identified during proposal planning and their corresponding mitigations.

The risk most worth naming explicitly is the third: this project’s own architecture (Figure 1) trains three modules on three different datasets and then combines them into one pipeline. Nothing about training each module well guarantees that its behavior transfers cleanly once it is applied to a different data source within the combined system. This is precisely why the integration plan (Section 5.7) is scoped as a required, separate evaluation step rather than an optional final check.

7 Ethical Considerations and Data Privacy

7.1 Data Provenance and Licensing

All three datasets (Sections 4.1–4.3) are publicly available on Kaggle under their respective dataset licenses. No new personal data collection is planned; all candidate and posting data used is pre-existing, de-identified at the level provided by the original dataset publishers. The project plans to use these datasets strictly for academic, non-commercial coursework, consistent with their stated terms.

7.2 Bias and Fairness

Both classification modules (genuineness and seniority) carry a fairness risk worth naming at the design stage rather than discovering after training: a model trained on historical postings can learn and reproduce whatever biases are already present in that historical data – for example, if certain job categories or phrasings are disproportionately associated with the fraud label in EMSCAD for reasons unrelated to actual fraud risk, a classifier could learn a spurious shortcut rather than a genuine signal. The evaluation

plan (Section 5.6) reports per-class metrics (not just aggregate accuracy) specifically so that uneven performance across classes – a symptom of exactly this kind of learned bias – would be visible rather than averaged away.

7.3 Explainability as an Ethical Requirement, Not Only a Technical One

This project’s emphasis on explainable recommendations (Section 5.5) is motivated by more than usability. A candidate who is shown a job recommendation with no rationale has no way to assess whether the system is matching them on relevant qualifications or on some other, potentially inappropriate signal. Surfacing which specific skills or requirements drove a match is planned as a way to make that assessment possible for the end user, not only as a debugging aid for the development team.

7.4 Limitations of This Section

This proposal is a coursework project evaluated on a fixed academic timeline (Table 4), using historical, de-identified public datasets rather than live user data. A production deployment of a system like this would need considerably more rigorous fairness auditing, ongoing bias monitoring, and explicit user consent mechanisms than this project’s scope allows; this section names the considerations relevant at proposal stage, not a complete responsible-deployment plan.

8 Team Responsibilities and Project Timeline

8.1 Team Responsibilities

Team Member	Reg. No.	Primary Responsibility
Himnish Kumar R	25MCAR0032	System design and architecture; overall pipeline design (Figure 1); Module 1 and Module 2 methodology planning
Gracy Sweetty T	25MCAR0031	Dataset identification and validation across all three sources; documentation
Rakshitha V	25MCAR0038	Data preprocessing pipeline design (text cleaning, TF-IDF/embedding strategy)
Ashish UK	25MCAR0027	Module 3 methodology planning; report compilation and literature review

Note: this allocation reflects primary ownership for planning purposes; all members contribute to review and documentation across sections.

8.2 Collaboration Plan

The team plans weekly check-ins aligned with the milestones in Table 4, so that dependencies between roles are surfaced early – for example, Module 2’s design (owned by Himnish Kumar R) depends on the LinkedIn dataset’s skill-taxonomy granularity assessment (owned by Gracy Sweetty T), and both need to be resolved before Module 3’s design (owned by Ashish UK) can finalize its planned candidate-side feature set. Shared documentation is planned to be maintained continuously through the phase rather than assembled only at submission time, so that this report reflects work actually completed at each milestone rather than a retrospective reconstruction.

8.3 Project Timeline

Table 4 sets out the planned schedule for this phase of the project, from initial problem selection through proposal submission.

Week	Milestone	Primary Owner(s)
Week 1	Problem ideation, real-world NLP problem selection, faculty consultation and approval	Full team
Week 2	Dataset identification and short-listing across candidate sources (Kaggle searches for fraud-detection, job-classification, and résumé-matching datasets)	Gracy Sweety T
Week 3	Literature review – collecting and synthesizing papers on job posting classification and personalized recommendation	Ashish UK, with review from full team
Week 4	Dataset validation (schema review, size/quality checks per dataset) and preprocessing pipeline design	Rakshitha V
Week 5	System architecture design, module-by-module methodology planning, and integration plan	Himnish Kumar R
Week 6	Proposal report drafting, internal review, and submission	Full team

Table 4: Planned six-week timeline for Phase 1 (proposal and system design).

This timeline is scoped to Phase 1 activities only – problem framing, dataset selection, and system design – consistent with this report’s stated scope (Section 2). Implementation, model training, and evaluation are planned for the project’s next phase and are outside this timeline’s scope.

Glossary of Key Terms

TF-IDF Term Frequency-Inverse Document Frequency: a text representation that weights each word in a document by how frequent it is locally versus how common it is across the whole corpus, giving higher weight to distinctive terms.

SMOTE Synthetic Minority Oversampling Technique: a method for addressing class imbalance by generating synthetic examples of the minority class rather than simply duplicating existing ones.

Class-weighted loss

A training adjustment that penalizes misclassifying a rare class more heavily than a common one, without generating any synthetic data.

Macro-F1 The F1 score (harmonic mean of precision and recall) averaged equally across all classes, regardless of how many examples each class has – as opposed to weighted-F1, which weights each class by its frequency.

R^2 (coefficient of determination)

A regression metric measuring the proportion of variance in the target variable explained by a model’s predictions; ranges up to 1.0, with higher values indicating a better fit.

Genuineness classification

This project's term for the binary task of labeling a job posting as legitimate or fraudulent.

Cold-start problem

The general recommendation-systems challenge of producing useful recommendations for a new user or item with little to no historical interaction data available.

Explainability

The property of a model's output being accompanied by a human-interpretable rationale for why that output was produced, as opposed to a bare score or ranking with no accompanying justification.

Content-based recommendation

A recommendation approach that ranks items by similarity between item content (e.g. job requirements) and user content (e.g. résumé text), as opposed to collaborative filtering, which relies on patterns across many users' historical interactions.

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